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United States Patent	12387785
Kind Code	B2
Date of Patent	August 12, 2025
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Structures for three-terminal memory cells

Abstract

The disclosed subject matter relates generally to structures for use in memory devices. More particularly, the present disclosure relates to three terminal resistive random-access (ReRAM) memory structures having source, drain, and control electrodes. The present disclosure provides a memory structure including a source electrode, a drain electrode, a control electrode laterally between the source electrode and the drain electrode, a hole generating layer above the control electrode, a dielectric channel layer above the hole generating layer, the dielectric channel layer contacts the source electrode and the drain electrode, a first spacer layer on a first side of the control electrode, and a second spacer layer on a second side of the control electrode. The first spacer layer and the second spacer layer isolate the source electrode and the drain electrode from the control electrode and the hole generating layer.

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Appl. No.: 18/065046

Filed: December 13, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240194255 A1	Jun. 13, 2024

Publication Classification

Int. Cl.: G11C13/00 (20060101); H01L23/522 (20060101); H10B99/00 (20230101); H10D64/23 (20250101)

U.S. Cl.:

CPC G11C13/0007 (20130101); H01L23/5226 (20130101); H10B99/00 (20230201); H10D64/251 (20250101); G11C2213/34 (20130101); G11C2213/52 (20130101)

Field of Classification Search

CPC: H10B (99/00)

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Background/Summary

FIELD OF THE INVENTION

(1) The disclosed subject matter relates generally to structures for use in memory devices. More particularly, the present disclosure relates to three terminal resistive random-access (ReRAM) memory structures having source, drain, and control electrodes.

BACKGROUND

(2) Semiconductor devices and integrated circuit (IC) chips have found numerous applications in the fields of physics, chemistry, biology, computing, and memory devices. An example of a memory device is a non-volatile (NV) memory device. NV memory devices are programmable and

have been extensively used in electronic products due to their ability to retain data for long periods, even after the power has been turned off. Exemplary categories for NV memory may include resistive random-access memory (ReRAM), erasable programmable read-only memory (EPROM), flash memory, ferroelectric random-access memory (FeRAM), and magnetoresistive random-access memory (MRAM).

(3) A ReRAM device includes a switching layer that is positioned between a bottom electrode and a top electrode. The ReRAM device can be programmed by changing the resistance across the switching layer to provide different content-storage states, namely a high-resistance state (HRS) and a low-resistance state (LRS), representing the stored bits of data. The switching layer can be modified by applying a programming voltage sufficient to create one or more conductive filaments bridging across the thickness of the switching layer, which sets the low-resistance state. The conductive filaments may be formed, for example, by the diffusion of a conductive species (e.g., metal ions) from one or both of the electrodes into the switching layer. The conductive filaments can be destroyed, also by the application of a programming voltage, to reset the resistive memory element to the high-resistance state. The content-storage state can be read by measuring a voltage drop across the resistive memory element after it is programmed.

SUMMARY

(4) In an aspect of the present disclosure, there is provided a memory structure including a source electrode, a drain electrode, a control electrode laterally between the source electrode and the drain electrode, the control electrode having a first side and a second side, a hole generating layer above the control electrode, a dielectric channel layer above the hole generating layer, the dielectric channel layer contacts the source electrode and the drain electrode, a first spacer layer on the first side of the control electrode, and a second spacer layer on the second side of the control electrode, in which the first spacer layer and the second spacer layer isolate the source electrode and the drain electrode from the control electrode and the hole generating layer.

(5) In another aspect of the present disclosure, there is provided a memory structure including a source electrode having an upper surface, a drain electrode having an upper surface, a control electrode laterally between the source electrode and the drain electrode, the control electrode has a lower surface, a first side, and a second side, a hole generating layer on the control electrode, a dielectric channel layer on the hole generating layer, the dielectric channel layer having a first side and a second side, in which the first side contacts the source electrode and the second side contacts the drain electrode, a first spacer layer on the first side of the control electrode, a second spacer layer on the second side of the control electrode, in which the first spacer layer and the second spacer layer isolate the source electrode and the drain electrode from the control electrode and the hole generating layer, a first interconnect structure below and in direct contact with the lower surface of the control electrode, a second interconnect structure above and in direct contact with the upper surface of the source electrode, and a third interconnect structure above and in direct contact with the upper surface of the drain electrode.

(6) In yet another aspect of the present disclosure, there is provided a memory structure including a first dielectric region, a first interconnect structure in the first dielectric region, a second dielectric region on the first dielectric region, a source electrode in the second dielectric region, a drain electrode in the second dielectric region, a control electrode in the second dielectric region, the control electrode is laterally between the source electrode and the drain electrode, the control electrode has a lower surface, a first side, and a second side, wherein the lower surface of the control electrode is on the first interconnect structure, a hole generating layer in the second dielectric region, the hole generating layer is on the control electrode, a dielectric channel layer in the second dielectric region, the dielectric channel layer is on the hole generating layer, the dielectric channel layer having a first side and a second side, wherein the first side contacts the source electrode and the second side contacts the drain electrode, a first spacer layer in the second dielectric region, the first spacer layer is on the first side of the control electrode, a second spacer

layer in the second dielectric region, the second spacer layer is on the second side of the control electrode, wherein the first spacer layer and the second spacer layer isolate the source electrode and the drain electrode from the control electrode and the hole generating layer, a third dielectric region on the second dielectric region, a second interconnect structure in the third dielectric region, the second interconnect structure is on the source electrode, and a third interconnect structure in the third dielectric region, the third interconnect structure is on the drain electrode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure may be understood by reference to the following description taken in conjunction with the accompanying drawings.
- (2) For simplicity and clarity of illustration, the drawings illustrate the general manner of construction, and certain descriptions and details of features and techniques may be omitted to avoid unnecessarily obscuring the discussion of the described embodiments of the present disclosure. Additionally, elements in the drawings are not necessarily drawn to scale. For example, the dimensions of some of the elements in the drawings may be exaggerated relative to other elements to help improve understanding of embodiments of the present disclosure. The same reference numerals in different drawings denote the same elements, while similar reference numerals may, but do not necessarily, denote similar elements.
- (3) FIG. 1 and FIG. 2 are cross-sectional views of example memory structures.
- (4) FIG. 3 through FIG. 11 are cross-sectional views depicting structures at successive fabrication stages of a processing method for fabricating an exemplary memory structure shown in FIG. 1.
- (5) FIG. 3 through FIG. 7 and FIG. 12 through FIG. 14 are cross-sectional views depicting structures at successive fabrication stages of a processing method for fabricating another exemplary memory structure shown in FIG. 2.

DETAILED DESCRIPTION

- (6) Various illustrative embodiments of the present disclosure are described below. The embodiments disclosed herein are exemplary and not intended to be exhaustive or limiting to the present disclosure.
- (7) Referring to FIG. 1 and FIG. 2, the example memory structures **100** may include a source electrode **110**, a drain electrode **112**, a control electrode **118** laterally between the source electrode **110** and the drain electrode **112**, a hole generating layer **116** above the control electrode **118**, a dielectric channel layer **114** above the hole generating layer **116**. The dielectric channel layer **114** may contact or directly contact the source electrode **110** and the drain electrode **112**. In some implementations, the hole generating layer **116** may be on or directly on the control electrode **118**, and the dielectric channel layer **114** may be on or directly on the hole generating layer **116**. The source electrode **110**, the drain electrode **112**, the dielectric channel layer **114**, the hole generating layer **116**, and the control electrode **118** may provide a memory cell, or a single memory cell unit. The memory structures **100** described herein may be formed above a substrate (not shown).
- (8) The control electrode **118** may have a first side **118a** and a second side **118b**. The first side **118a** of the control electrode **118** may be oppositely facing the second side **118b** of the control electrode **118**. The control electrode **118** may have a lower surface **118c**. The lower surface **118c** may adjoin the first side **118a** and the second side **118b**. The hole generating layer **116** may have a first side **116a** and a second side **116b**. A first spacer layer **134** may cover at least the first side **118a** of the control electrode **118** and the first side **116a** of the hole generating layer **116**. A second spacer layer **136** may cover at least the second side **118b** of the control electrode **118** and the second side **116b** of the hole generating layer **116**. The first spacer layer **134** and the second spacer layer **136** may isolate the source electrode **110** and the drain electrode **112** from the control electrode **118** and the

hole generating layer **116**. For example, the first spacer layer **134** may be on or directly on the first side **118a** of the control electrode **118** and the first side **116a** of the hole generating layer **116** while the second spacer layer **136** may be on or directly on the second side **118b** of the control electrode **118** and the second side **116b** of the hole generating layer **116**.

(9) The dielectric channel layer **114** may have a first side **114a** and a second side **114b**. The first side **114a** may contact or directly contact the source electrode **110** and the second side **114b** may contact or directly contact the drain electrode **112**. The first side **114a** of the dielectric channel layer **114** may have an upper portion **111** and a lower portion **113**. The second side **114b** of the dielectric channel layer **114** may have an upper portion **115** and a lower portion **117**. The first spacer layer **134** may overlap with the lower portion **113** of the first side **114a** of the dielectric channel layer **114**. The second spacer layer **136** may overlap with the lower portion **117** of the second side **114b** of the dielectric channel layer **114**. For example, the first spacer layer **134** may be on or directly on the lower portion **113** of the first side **114a** of the dielectric channel layer **114**. The second spacer layer **136** may be on or directly on the lower portion **117** of the second side **114b** of the dielectric channel layer **114**. The source electrode **110** may contact or directly contact the upper portion **111** of the first side **114a** of the dielectric channel layer **114**. The drain electrode **112** may contact or directly contact the upper portion **115** of the second side **114b** of the dielectric channel layer **114**.

(10) The source electrode **110** may have an upper surface **110t**. The drain electrode **112** may have an upper surface **112t**. The dielectric channel layer **114** may have an upper surface **114t**. The upper surface **110t** of the source electrode **110**, the upper surface **112t** of the drain electrode **112**, and the upper surface **114t** of the dielectric channel layer **114** may be substantially coplanar with each other. For example, the upper surface **114t** of the dielectric channel layer **114** may be substantially coplanar with the upper surface **110t** of the source electrode **110** and the upper surface **112t** of the drain electrode **112**.

(11) Interconnect structures, such as vias, landing pads, and conductive lines, may be formed above or below the control electrode **118**, the source electrode **110**, and the drain electrode **112**. These interconnect structures may provide routing or wiring of electrical signals and may connect various devices or components within an IC chip to perform desired functions. As shown, the exemplary memory structures **100** may further include a first interconnect structure **122** below the control electrode **118**, a second interconnect structure **120** above the source electrode **110**, and a third interconnect structure **124** above the drain electrode **112**. A conductive line **126** may be on the second interconnect structure **120** and a conductive line **130** may be on the third interconnect structure **124**. In some implementations, the first interconnect structure **122** may contact or directly contact the lower surface **118c** of the control electrode **118**. The second interconnect structure **120** may contact or directly contact the upper surface **110t** of the source electrode **110**. The third interconnect structure **124** may contact or directly contact the upper surface **112t** of the drain electrode **112**. In the example shown in FIG. 1, the first interconnect structure **122** may be a via. The first interconnect structure **122** may be coupled and disposed on a conductive line **129**. In the example shown in FIG. 2, the first interconnect structure **122** may be a landing pad. Alternatively, the first interconnect structure **122** shown in FIG. 1 may be a landing pad, and the first interconnect structure **122** shown in FIG. 2 may be a via, in which the first interconnect structure **122** may be coupled and disposed on a conductive line.

(12) In the example shown in FIG. 2, the second interconnect structure **120** may overlap with the source electrode **110** and the dielectric channel layer **114** such that the second interconnect structure **120** may contact or directly contact the upper surface **110t** of the source electrode **110** and the upper surface **114t** of the dielectric channel layer **114**. The third interconnect structure **124** may overlap with the drain electrode **112** and the dielectric channel layer **114** such that the third interconnect structure **124** may contact or directly contact the upper surface **112t** of the drain electrode **112** and the upper surface **114t** of the dielectric channel layer **114**. Alternatively, in the example shown in FIG. 1, the second interconnect structure **120** may only contact or directly

contact the upper surface **110t** of the source electrode **110**, and the third interconnect structure **124** may only contact or directly contact the upper surface **112t** of the drain electrode **112**.

(13) Advantageously, having the hole generating layer **116** above the control electrode **118** and the dielectric channel layer **114** above the hole generating layer **116** may enable the first interconnect structure **122** to be positioned below the control electrode **118** while the second interconnect structure **120** and the third interconnect structure **124** to be positioned above the source electrode. The first interconnect structure **122** may also be formed in a different dielectric region from the second interconnect structure **120** and the third interconnect structure **124**. The lateral distance between the second interconnect structure **120** and the third interconnect structure **124** can be reduced since the first interconnect structure **122** connecting to the control electrode does not need to be arranged laterally between the second interconnect structure **120** and the third interconnect structure **124** and does not need to be in the same dielectric region and the second interconnect structure **120** and the third interconnect structure **124**. The memory structures **100** described herein can achieve a smaller size when compared to other memory structures where an interconnect structure is arranged laterally between the second interconnect structure and the third interconnect structure **124**.

(14) The exemplary memory structures **100** may also include a first dielectric region **102**, a second dielectric region **104** on or directly on the first dielectric region **102**, and a third dielectric region **106** on or directly on the second dielectric region **104**. The dielectric regions **102**, **104**, **106** may be a region formed by the middle of line (MOL) or back end of line (BEOL) processing of an IC chip. The dielectric regions **102**, **104**, **106** may include a metallization level. The first interconnect structure **122** may be in the first dielectric region **102**. The source electrode **110**, the drain electrode **112**, the control electrode **118**, the hole generating layer **116**, the dielectric channel layer **114**, the first spacer layer **134**, and the second spacer layer **136** may be in the second dielectric region **104**. The second interconnect structure **120**, the third interconnect structure **124**, and the conductive lines **126**, **130** may be in the third dielectric region **106**.

(15) The source electrode **110**, the drain electrode **112**, and the control electrode **118** may include a metal. Examples of the metal for the source electrode **110**, the drain electrode **112**, and the control electrode **118** may include, but are not limited to, tungsten, molybdenum, vanadium, strontium, cobalt, tantalum, titanium, hafnium, copper, aluminum, or an alloy thereof. In an embodiment, the electrodes **110**, **112**, **118** may have the same material. In another embodiment, the source electrode **110**, the drain electrode **112**, and the control electrodes **118** have different materials from each other. In yet another embodiment, the source electrode **110** and the drain electrode **112** may have the same material while the control electrode **118** may have a different material from the source electrode **110** and the drain electrode **112**.

(16) The dielectric channel layer **114** may include an oxide. In some embodiments, the dielectric channel layer **114** may include an oxide of the metal in the source electrode **110** and the drain electrode **112**. In other embodiments, the dielectric channel layer **114** may include an oxide of tungsten, molybdenum, vanadium, or strontium cobalt alloy. The hole generating layer **116** may include a material saturated with hydrogen. For example, the material may have a crystalline structure and have hydrogen gas intercalated at the interstitial defect sites in the material. The hole generating layer **116** may include a hydrogen doped dielectric material or a hydrogen doped metal. For example, the hole generating layer **116** may include hydrogen doped silicon dioxide (SiO_2 —H), platinum hydride (Pt—H), or palladium hydride (Pd—H). The hole generating layer **116** may be capable of providing holes (i.e., positive charge carriers or protons) that migrate from the hole generating layer **116** towards the dielectric channel layer **114** under the influence of an electric field.

(17) As an illustrative example, during the operation of the memory structures **100** shown in FIG. 1 and FIG. 2, a bias voltage may be applied to the control electrode **118** through the first interconnect structure **122**. The number of holes migrated towards the dielectric channel layer **114** may be

dependent on the magnitude and direction of the bias voltage. A current may be allowed to flow through the dielectric channel layer **114** and between the source electrode **110** and the drain electrode **112**. The migration of holes in and out of the dielectric channel layer **114** may increase or decrease the conductance of the dielectric channel layer **114**, thereby increasing or decreasing the current flow between the source electrode **110** and the drain electrode **112**. Thus, the control of the bias voltage applied to the control electrode **118** may modulate the conductance of the dielectric channel layer **114**.

(18) The dielectric regions **102**, **104**, **106** may include dielectric materials **108**, **128**, **132**, respectively. Examples of the dielectric materials **108**, **128**, **132** may include, but are not limited to, silicon dioxide (SiO_2), silicon oxynitride (SiON), silicon nitride (Si_3N_4), Nitrogen doped silicon carbide (SiCN), SiC_xH_y (i.e., BLOK^{TM}), or $\text{SiN}_w\text{C}_x\text{H}_y$ (i.e., NBLoK^{TM}), wherein each of w , x , y , and z independently has a value greater than 0 and less than 0.75, tetraethyl orthosilicate (TEOS), or a material having a chemical composition of $\text{SiC}_x\text{O}_y\text{H}_z$, wherein x , y , and z are in stoichiometric ratio.

(19) The spacer layers **134**, **136** may include, but are not limited to, silicon dioxide (SiO_2), silicon oxynitride (SiON), silicon nitride (Si_3N_4), Nitrogen doped silicon carbide (SiCN), SiC_xH_y (i.e., BLOK^{TM}), or $\text{SiN}_w\text{C}_x\text{H}_y$ (i.e., NBLoK^{TM}), wherein each of w , x , y , and z independently has a value greater than 0 and less than 0.75, tetraethyl orthosilicate (TEOS), or a material having a chemical composition of $\text{SiC}_x\text{O}_y\text{H}_z$, wherein x , y , and z are in stoichiometric ratio. The spacer layers **134**, **136** may serve to prevent electrical shorts or migration of holes between the source and drain electrodes **110**, **112** and the sides **116a**, **116b** of the hole generating layer **116** and the sides **118a**, **118b** of the control electrode **118**, and thereby ensuring that the migration of holes occurs only between the horizontal segment **116c** of the hole generating layer **116** and the horizontal segment **114c** of the dielectric channel layer **114**.

(20) FIGS. **3** through **11** show structures at successive fabrication stages of a processing method for fabricating an exemplary memory structure shown in FIG. **1**.

(21) As used herein, “deposition techniques” refer to the process of applying a material over another material (or the substrate). Exemplary techniques for deposition include, but are not limited to, spin-on coating, sputtering, chemical vapor deposition (CVD), physical vapor deposition (PVD), molecular beam deposition (MBD), pulsed laser deposition (PLD), liquid source misted chemical deposition (LSMCD), atomic layer deposition (ALD).

(22) Additionally, “patterning techniques” include deposition of material or photoresist, patterning, exposure, development, etching, cleaning, and/or removal of the material or photoresist as required in forming a described pattern, structure, or opening. Examples of techniques for patterning include, but are not limited to, wet etch lithographic processes, dry etch lithographic processes, or direct patterning processes. Such techniques may use mask sets and mask layers.

(23) Referring to FIG. **3**, a first dielectric region **102** may be formed over a substrate (not shown). The first dielectric region **102** may include a first interconnect structure **122** formed in a dielectric material **108**. A control electrode **118** may be formed on the dielectric material **108** and the first interconnect structure **122**, for example, using deposition techniques. A hole generating layer **116** may be formed on the control electrode **118**, for example, using deposition techniques. In an example, the hole generating layer **116** may be formed by depositing a silicon dioxide layer. The silicon dioxide layer may be injected, doped, or sputtered by a forming gas **138**, such as hydrogen gas. In another example, the hole generating layer **116** may be formed by depositing platinum or palladium. The deposited platinum or palladium may be injected, doped, or sputtered by the forming gas **138** such that the platinum or palladium is saturated with the forming gas.

(24) Referring to FIG. **4**, a dielectric channel layer **114** may be formed on the hole generating layer **116** using deposition techniques. A mask layer **140** may be formed on the dielectric channel layer **114** and then patterned using patterning techniques. The mask layer **140** may be used for the subsequent patterning of the control electrode **118**, the hole generating layer **116**, and the dielectric

channel layer **114**. Referring to FIG. 5, the control electrode **118**, the hole generating layer **116**, and the dielectric channel layer **114** may be patterned using patterning techniques so that the patterned control electrode **118**, hole generating layer **116**, and dielectric channel layer **114** are aligned vertically over the first interconnect structure **122**. The control electrode **118** may be patterned to form sides **118a**, **118b**, the hole generating layer **116** may be patterned to form sides **116a**, **116b**, and the dielectric channel layer **114** may be patterned to form sides **114a**, **114b**. The dielectric channel layer **114** may also have an upper surface **114t**. The control electrode **118** may be patterned such that it has a lower surface **118c** that contacts or directly contacts the first interconnect structure **122**.

(25) Referring to FIG. 6, a spacer material layer **135** may be formed using a deposition technique, and preferably, using conformal deposition (e.g., ALD) such that the spacer material layer **135** covers the sides **118a**, **118b** of the control electrode, the sides **116a**, **116b** of the hole generating layer **116**, and the sides **114a**, **114b** and the upper surface **114t** of the dielectric channel layer **114**. The spacer material layer **135** may also cover the dielectric material **108** in the first dielectric region **102**.

(26) Referring to FIG. 7, the spacer material layer **135** may be etched using a directional etching, such as anisotropic etching, to form a first spacer layer **134** and a second spacer layer **136**. The etching of the spacer material layer **135** may be controlled such that the first spacer layer **134** and the second spacer layer **136** cover the sides **118a**, **118b** of the control electrode and the sides **116a**, **116b** of the hole generating layer **116**. Additionally, a lower portion **113** of side **114a** and a lower portion **117** of side **114b** may be covered by the first spacer layer **134** and the second spacer layer **136**, respectively, while an upper portion **111** of side **114a** and an upper portion **115** of side **114b** may be exposed and uncovered by the first spacer layer **134** and the second spacer layer **136**, respectively. The etching of the spacer material layer **135** may also expose the upper surface **114t** of the dielectric channel layer **114** and portions of the dielectric material **108** in the first dielectric region **102**.

(27) Referring to FIG. 8, a second dielectric region **104** may be formed on the first dielectric region **102**. The second dielectric region **104** may include a dielectric material **128** formed on the dielectric material **108** using deposition techniques. The dielectric material **128** may have an upper surface **128t**. The dielectric material **128** may be patterned using patterning techniques to form source/drain openings **146** in the second dielectric region **104**. The source/drain openings **146** may be defined in the dielectric material **128**.

(28) Referring to FIG. 9, a metal layer **142** may be formed in the source/drain openings **146**. The metal layer **142** may also be formed over the dielectric material **108**, the spacer layers **134**, **136**, the dielectric material **128**, and the dielectric channel layer **114** using deposition techniques. The metal layer **142** may be deposited in the source/drain openings **146** such that the metal layer **142** is on the upper portion **111** of the side **114a** of the dielectric channel layer **114**, the upper portion **115** of the side **114b** of the dielectric channel layer **114**. The metal layer **142** may also extend outside of the source/drain openings **146** and overlie the upper surface **128t** of the dielectric material **128** and the upper surface **114t** of the dielectric channel layer **114**.

(29) Referring to FIG. 10, a chemical mechanical planarization (CMP) process may be performed on the structure shown in FIG. 9 to remove portions of the metal layer **142** and expose the upper surface **114t** of the dielectric channel layer **114** and the upper surface **128t** of the dielectric material **128**. A source electrode **110** and a drain electrode **112** may be formed after the CMP process. The source electrode **110** may have an upper surface **110t** and the drain electrode **112** may have an upper surface **112t**. The CMP process may ensure that the upper surface **110t** of the source electrode **110**, and the upper surface **112t** of the drain electrode **112** are substantially coplanar with the upper surface **114t** of the dielectric channel layer **114** and the upper surface **128t** of the dielectric material **128**. In an implementation, the source electrode **110** and the drain electrode **112** may include the same metal. The second dielectric region **104** may include the source electrode

110, the drain electrode **112**, the spacer layers **134**, **136**, the control electrode **118**, the hole generating layer **116**, and the dielectric channel layer **114**, in addition to the dielectric material **128**. (30) Referring to FIG. **11**, a third dielectric region **106** may be formed on the second dielectric region **104**. The third dielectric region **106** may include a dielectric material **132** formed on the upper surface **110t** of the source electrode **110**, the upper surface **112t** of the drain electrode, the upper surface **114t** of the dielectric channel layer **114**, and the dielectric material **128** using the deposition techniques described herein. Interconnect openings **144** may be formed in the dielectric material **132** using patterning techniques. Interconnect structures, as described in FIG. **1**, may be formed in the interconnect openings **144**, for example, by forming a metal, such as cobalt (Co), copper (Cu), aluminum (Al), or an alloy thereof, in the interconnect openings **144**. Other suitable types of metals or alloys may also be useful. The interconnect openings **144** may be formed such that they are aligned vertically above the upper surface **110t** of the source electrode **110** and the upper surface **112t** of the drain electrode **112**.

(31) FIGS. **3** through **7** and FIGS. **12** through **14** show structures at successive fabrication stages of a processing method for fabricating an exemplary memory structure shown in FIG. **2**.

(32) Referring to FIG. **12** (FIG. **12** continues from the structure shown in FIG. **7**), a metal layer **142** may be formed over the dielectric material **108**, the spacer layers **134**, **136**, and the dielectric channel layer **114** using deposition techniques. The metal layer **142** may be deposited on the upper portion **111** of the side **114a** of the dielectric channel layer **114**, the upper portion **115** of the side **114b** of the dielectric channel layer **114**, and the upper surface **114t** of the dielectric channel layer **114**. The deposition of the metal layer **142** in FIG. **12** may deposit lesser material as compared to the deposition of the metal layer **142** in FIG. **9**.

(33) Referring to FIG. **13**, the metal layer **142** may be etched using a directional etching such as anisotropic etching. For example, horizontal segments of the metal layer **142** covering the dielectric material **108** and the upper surface **114t** of the dielectric channel layer **114** may be removed while vertical segments of the metal layer **142** covering the spacer layers **134**, **136** and the upper portions **111**, **115** of the sides **114a**, **114b** of the dielectric channel layer **114** are retained. A source electrode **110** and a drain electrode **112** may be formed after the etching of the metal layer **142**. The source electrode **110** may have an upper surface **110t** and the drain electrode **112** may have an upper surface **112t**. In an implementation, the source electrode **110** and the drain electrode **112** may include the same metal.

(34) Referring to FIG. **14**, a second dielectric region **104** may be formed on the first dielectric region **102**. The second dielectric region **104** may include a dielectric material **128** formed laterally adjacent to the source electrode **110** and the drain electrode **112** and on the dielectric material **108** using deposition techniques. Although not shown, the dielectric material **128** may be deposited over the upper surface **114t** of the dielectric channel layer **114**, the upper surface **110t** of the source electrode **110**, and the upper surface **112t** of the drain electrode **112**. A CMP process may be performed on the deposited dielectric material **128** to expose the upper surface **114t** of the dielectric channel layer **114**, the upper surface **110t** of the source electrode **110**, and the upper surface **112t** of the drain electrode **112**. The CMP process may also ensure that the upper surface **110t** of the source electrode **110**, the upper surface **112t** of the drain electrode **112** are substantially coplanar with the upper surface **114t** of the dielectric channel layer **114**. The second dielectric region **104** may include the source electrode **110**, the drain electrode **112**, the spacer layers **134**, **136**, the control electrode **118**, the hole generating layer **116**, and the dielectric channel layer **114**.

(35) A third dielectric region **106** may be formed on the second dielectric region **104**. The third dielectric region **106** may include a dielectric material **132** formed on the upper surface **110t** of the source electrode **110**, the upper surface **112t** of the drain electrode, the upper surface **114t** of the dielectric channel layer **114**, and the dielectric material **128** using the deposition techniques described herein. Interconnect openings **144** may be formed in the dielectric material **132** using patterning techniques. Interconnect structures, as described in FIG. **1**, may be formed in the

interconnect openings **144**, for example, by forming a metal, such as cobalt (Co), copper (Cu), aluminum (Al), or an alloy thereof, in the interconnect openings **144**. Other suitable types of metals or alloys may also be useful. The interconnect openings **144** may be formed such that they are aligned vertically above the upper surface **110t** of the source electrode **110** and the upper surface **112t** of the drain electrode **112**. The vertical alignment of the interconnect openings **144** may also overlap with the source electrode **110** and the dielectric channel layer **114** and the drain electrode **112** and the dielectric channel layer **114**.

(36) Throughout this disclosure, it is to be understood that if a method is described herein as involving a series of steps, the order of such steps as presented herein is not necessarily the only order in which such steps may be performed, and certain of the stated steps may possibly be omitted and/or certain other steps not described herein may possibly be added to the method. Furthermore, the terms “comprise”, “include”, “have”, and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or device that comprises a list of elements is not necessarily limited to those elements, but may include other elements not expressly listed or inherent to such process, method, article, or device. Occurrences of the phrase “in an embodiment” herein do not necessarily all refer to the same embodiment.

(37) The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein. Furthermore, there is no intention to be bound by any theory presented in the preceding background or the following detailed description. Additionally, the various tasks and processes described herein may be incorporated into a more comprehensive procedure or process having additional functionality not described in detail herein.

(38) References herein to terms modified by language of approximation, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. The language of approximation may correspond to the precision of an instrument used to measure the value and, unless otherwise dependent on the precision of the instrument, may indicate $\pm 10\%$ of the stated value(s).

(39) As will be readily apparent to those skilled in the art upon a complete reading of the present application, the disclosed semiconductor devices and methods of forming the same may be employed in manufacturing a variety of different integrated circuit products, including, but not limited to, memory cells, NV memory devices, FinFET transistor devices, CMOS devices, etc.

Claims

1. A memory structure comprising: a source electrode; a drain electrode; a control electrode laterally between the source electrode and the drain electrode, the control electrode having a first side and a second side; a hole generating layer above the control electrode; a dielectric channel layer above the hole generating layer, the dielectric channel layer contacts the source electrode and the drain electrode; a first spacer layer on the first side of the control electrode; and a second spacer layer on the second side of the control electrode, wherein the first spacer layer and the second spacer layer isolate the source electrode and the drain electrode from the control electrode and the hole generating layer.
2. The memory structure of claim 1, wherein the dielectric channel layer has a first side and a second side, the first side contacts the source electrode, and the second side contacts the drain electrode.

3. The memory structure of claim 2, further comprising: a first interconnect structure below the control electrode; a second interconnect structure above the source electrode; and a third interconnect structure above the drain electrode.
4. The memory structure of claim 2, wherein the first spacer layer overlaps with a lower portion of the first side of the dielectric channel layer, and the second spacer layer overlaps with a lower portion of the second side of the dielectric channel layer.
5. The memory structure of claim 4, wherein the source electrode contacts an upper portion of the first side of the dielectric channel layer and the drain electrode contacts an upper portion of the second side of the dielectric channel layer.
6. The memory structure of claim 3, wherein the second interconnect structure overlaps with the source electrode and the dielectric channel layer, and the third interconnect structure overlaps with the drain electrode and the dielectric channel layer.
7. The memory structure of claim 6, further comprising: a first dielectric region, wherein the first interconnect structure is in the first dielectric region; and a second dielectric region on the first dielectric region, wherein the source electrode, the drain electrode, the control electrode, the dielectric channel layer, the hole generating layer, the first spacer layer, and the second spacer layer are in the second dielectric region.
8. The memory structure of claim 7, further comprising a third dielectric region on the second dielectric region, wherein the second interconnect structure and the third interconnect structure are in the third dielectric region.
9. The memory structure of claim 1, wherein the source electrode has an upper surface, the drain electrode has an upper surface, and the dielectric channel layer has an upper surface, and wherein the upper surface of the source electrode and the upper surface of the drain electrode are substantially coplanar with the upper surface of the dielectric channel layer.
10. The memory structure of claim 1, wherein the hole generating layer includes a hydrogen doped dielectric material or a hydrogen doped metal.
11. The memory structure of claim 10, wherein the hole generating layer includes hydrogen doped silicon dioxide, platinum hydride, or palladium hydride.
12. The memory structure of claim 1, wherein the dielectric channel layer includes an oxide of tungsten, molybdenum, vanadium, or strontium cobalt alloy.
13. A memory structure comprising: a source electrode having an upper surface; a drain electrode having an upper surface; a control electrode laterally between the source electrode and the drain electrode, the control electrode has a lower surface, a first side, and a second side; a hole generating layer on the control electrode; a dielectric channel layer on the hole generating layer, the dielectric channel layer having a first side and a second side, wherein the first side contacts the source electrode and the second side contacts the drain electrode; a first spacer layer on the first side of the control electrode; a second spacer layer on the second side of the control electrode, wherein the first spacer layer and the second spacer layer isolate the source electrode and the drain electrode from the control electrode and the hole generating layer; a first interconnect structure below and in contact with the lower surface of the control electrode; a second interconnect structure above and in contact with the upper surface of the source electrode; and a third interconnect structure above and in contact with the upper surface of the drain electrode.
14. The memory structure of claim 13, wherein the upper surface of the source electrode and the upper surface of the drain electrode are substantially coplanar with the upper surface of the dielectric channel layer.
15. The memory structure of claim 13, wherein the first spacer layer overlaps with a lower portion of the first side of the dielectric channel layer, and the second spacer layer overlaps with a lower portion of the second side of the dielectric channel layer.
16. The memory structure of claim 15, wherein the source electrode contacts an upper portion of the first side of the dielectric channel layer and the drain electrode contacts an upper portion of the

second side of the dielectric channel layer.

17. The memory structure of claim 16, wherein the second interconnect structure overlaps with the source electrode and the dielectric channel layer, and the third interconnect structure overlaps with the drain electrode and the dielectric channel layer.

18. The memory structure of claim 13, wherein the hole generating layer includes a hydrogen doped dielectric material or a hydrogen doped metal.

19. The memory structure of claim 18, wherein the hole generating layer includes hydrogen doped silicon dioxide, platinum hydride, or palladium hydride.

20. A memory structure comprising: a first dielectric region; a first interconnect structure in the first dielectric region; a second dielectric region on the first dielectric region; a source electrode in the second dielectric region; a drain electrode in the second dielectric region; a control electrode in the second dielectric region, the control electrode is laterally between the source electrode and the drain electrode, the control electrode has a lower surface, a first side, and a second side, wherein the lower surface of the control electrode is on the first interconnect structure; a hole generating layer in the second dielectric region, the hole generating layer is on the control electrode; a dielectric channel layer in the second dielectric region, the dielectric channel layer is on the hole generating layer, the dielectric channel layer having a first side and a second side, wherein the first side contacts the source electrode and the second side contacts the drain electrode; a first spacer layer in the second dielectric region, the first spacer layer is on the first side of the control electrode; a second spacer layer in the second dielectric region, the second spacer layer is on the second side of the control electrode, wherein the first spacer layer and the second spacer layer isolate the source electrode and the drain electrode from the control electrode and the hole generating layer; a third dielectric region on the second dielectric region; a second interconnect structure in the third dielectric region, the second interconnect structure is on the source electrode; and a third interconnect structure in the third dielectric region, the third interconnect structure is on the drain electrode.
